Deconvolution of Transcriptomic Data Shows Biologically and Clinically Relevant Signals in Pancreatic Tumors

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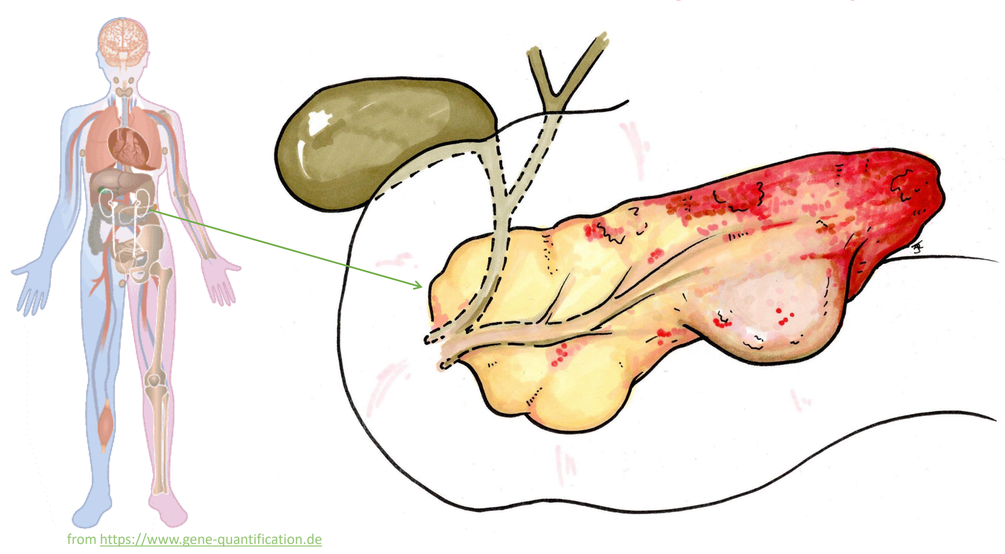
Pancreatic cancer is a significant **challenge** to oncology. The early stages of the disease are **asymptomatic**, which limits diagnosis and treatment of the neoplastic process and thus lead to bad survival prognosis [1,2].

- *Asymptomatic*
- *High progression rate*
- *Large social impact*
- *High metastasis rate*
- *High mortality*



Transcriptomics

Datasets (D):

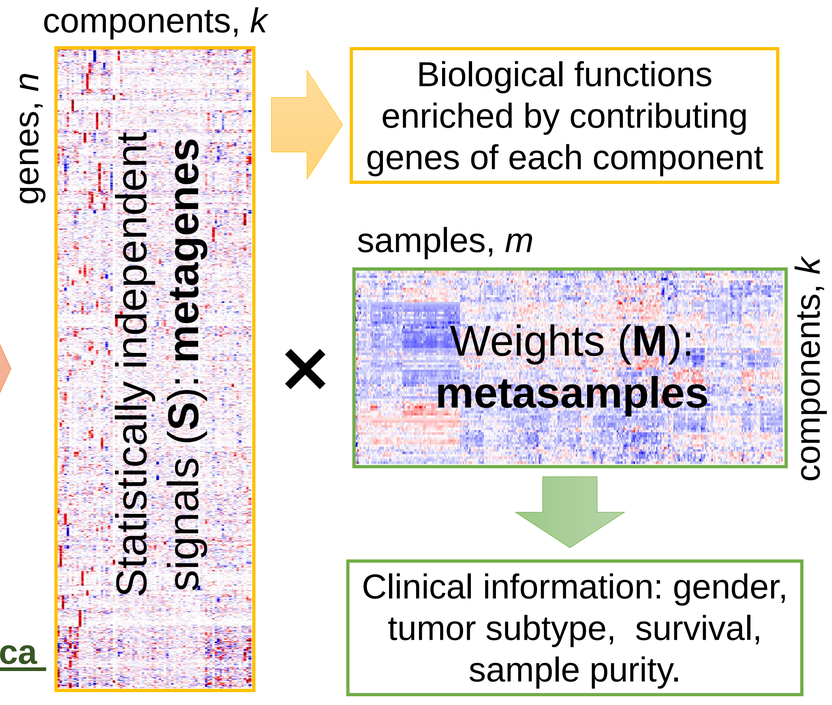
- **TCGA**: 183 samples
- **DKFZ**:[1] 457 samples
- **Bailey**:[2] 96 samples

ICA

https://gitlab.com/biomodlih/consica

Independent component analysis transforms the data into a matrix product of statistically independent transcriptional signals and their weight.

$$D_{nm} = S_{nk} \times M_{km}$$



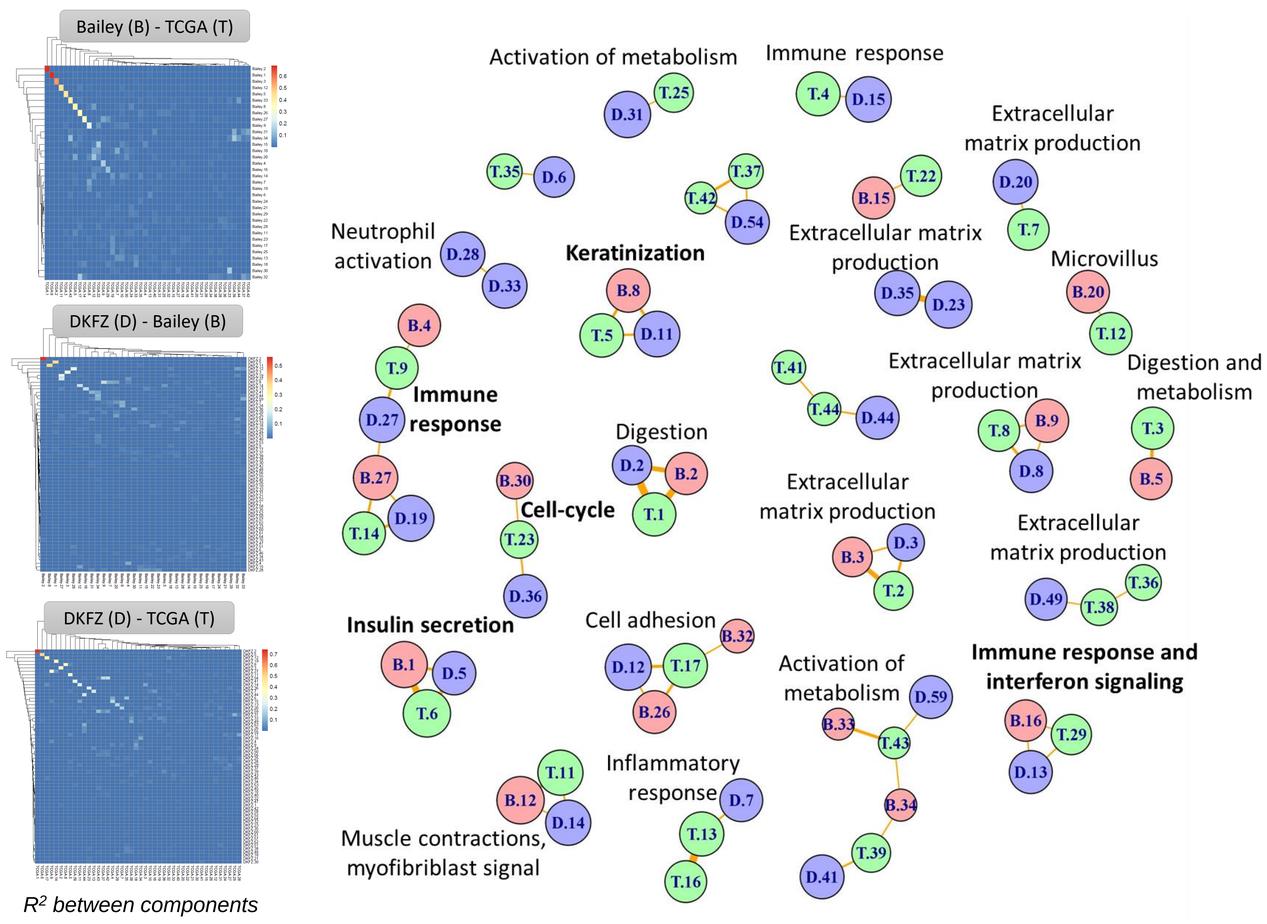
Research goals:

1. Identify **pathophysiological processes** affecting survival of patients with pancreatic cancer and determine the role of active signaling pathways.
2. Characterize **tumor purity** in unsupervised manner.
3. Predict survival of new patients.

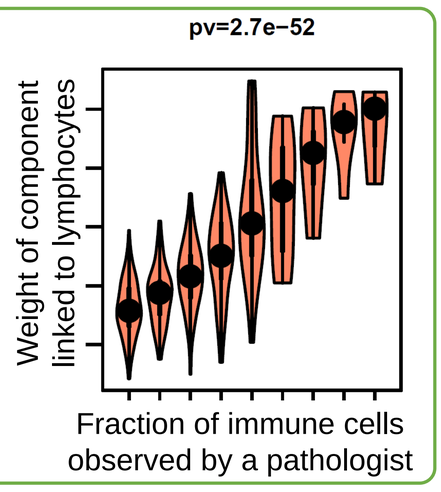
$$RS_j = \sum_{i=1}^{i=k} R_i^2 H_i M_{i,j}^*$$

Risk score (RS) is calculated as the weighted sum on scaled rows of ***M*** and Cox log hazard ratio (*H*). Stability of the components (R^2) is also considered.

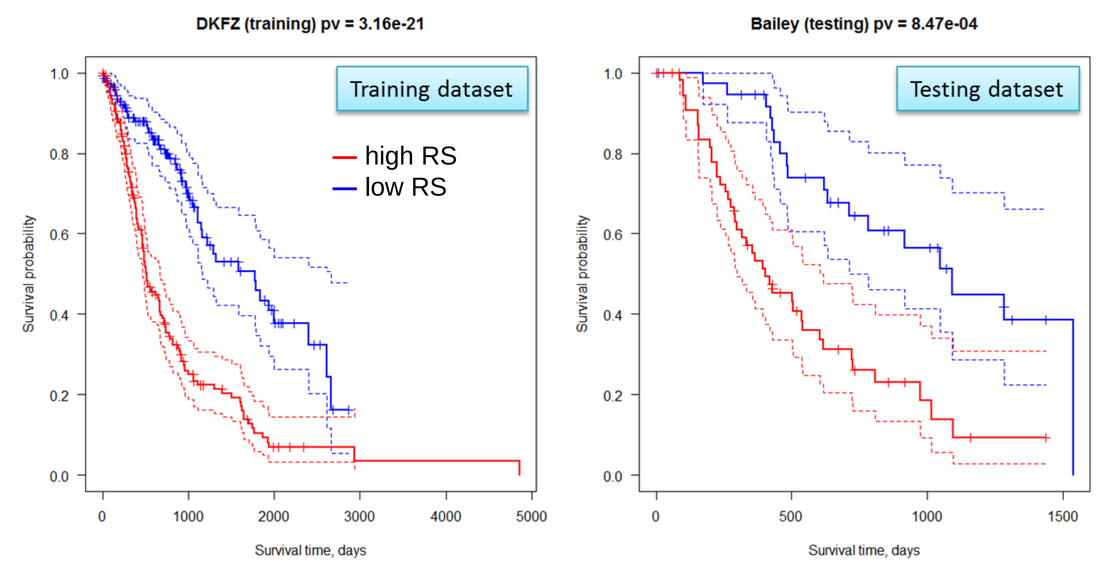
Three dataset were first analyzed independently and components were linked using R^2 between columns of **S** matrix. Many components were found reproducible between datasets and formed a graph presented below.



2 ICA was able to detect transcriptional signals specific to stroma and tumors. In DKFZ dataset, abundance of normal pancreas tissue as well as immune cells was detected in an unsupervised manner and correlated to an independent observation of an immuno-histopahthologist. For Bailey and TCGA dataset such correlation was observed with *in silico* predictions.



3 Here we combined DKFZ (training) and Bailey (testing) datasets. ICA was performed on the joint data. Risk scores (RS) were calculated as in [3] and visualized:



1 Components identified by ICA were annotated by biological functions (GO) and linked to survival using Cox regression as is described in [3].

Increased risk:
- **keratinization**
- **cell cycle**
- response to hypoxia
- neoangiogenesis
- activation of ERK-signaling

No effect:
- immune response
- gender
- axon development

Reduced risk:
- secretion activity (normal)
- digestion
- antigen binding

Unlike in melanoma [3], no direct link was found between immune response and survival: perhaps due to a dual / antinomic effect.

Conclusions:

► Pathophysiological **processes** that affect survival in patients with pancreatic cancer are **cell proliferation** and **keratinization**. No strong effect of immune components on survival was detected.

► **Tumor purity** was characterized. It strongly correlated with independent observations of immune cells in DKFZ cohort and in silico estimation in Bailey dataset.

►**Risk score** calculated for the testing dataset (Bailey) strongly correlated with the survival (p-value < 0.001).

References

1. Bauer et al., International Journal of Cancer, 2018, 142(5):1010-21
2. Bailey et al., Nature, 2016, doi:10.1038
3. Nazarov et al., BMC Genomics, 2019, 12(1):132

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